Salma Labidi

Data Science Engineer

Tunis, Tunisia | +216 56 788 168 | salmalabidi8@gmail.com

LinkedIn | GitHub | Portfolio

Profile

Data Science Engineer with hands-on experience in **Machine Learning, AI and MLOps**. Skilled in developing end-to-end ML solutions, predictive models and AI-powered applications, from data engineering and experimentation to deployment and monitoring. Interested in building scalable, production-ready AI systems.

Education

Engineering Degree in Computer Science – Data Science 2023–2026
ESPRIT

Bachelor's Degree in Computer Systems Engineering and Embedded Systems 2020–2023
Faculty of Sciences of Tunis (FST)

Professional Experience

Engineering Final-Year Internship (PFE) – Data Science & MLOps **Feb.–Aug. 2026**
Huawei (Client: Ooredoo Tunisia) | Tunis, Tunisia

- Designed an **end-to-end MLOps pipeline** for inbound and outbound roaming optimization, integrating partner quality forecasting, anomaly detection, revenue loss estimation and network diagnosis.
- Developed **LightGBM, Isolation Forest, One-Class SVM and XGBoost** models; orchestrated workflows with **Airflow, MLflow and PostgreSQL**. Built **Grafana dashboards**, drift monitoring, a **React/FastAPI** application and a private **RAG/ReAct assistant** using Ollama and LangGraph. Implemented traffic steering optimization with **HiGHS**, achieving an **11.4% improvement** in the evaluated optimization scenario; revenue prediction achieved a **0.905 TND MAE**.

Data Science Intern **Jul.–Aug. 2025**
ArmadaDelivery | Kuwait (Remote)

- Designed a **predictive geocoding system** converting noisy Arabic and English Kuwaiti addresses into GPS coordinates for last-mile delivery.
- Engineered an **NLP and geospatial pipeline** combining RapidFuzz, TF-IDF, multilingual Sentence-Transformer embeddings, KDE weighting and Geohash encoding.
- Developed a hybrid **FAISS nearest-neighbor search and XGBoost** architecture; deployed it as a **Python REST API** integrated with the Ettijahet Directions platform (Express.js) for geocoding, distance and ETA estimation.

Web Development Intern **Jul.–Aug. 2024**
KeyStone | Tunis, Tunisia

- Developed a **Symfony application** for security incident reporting and risk analysis, featuring evidence uploads, risk assessment, automated reports and interactive dashboards.

Bachelor's Final-Year Internship (PFE) **Feb.–May 2023**
SAGEM | Tunisia

- Automated a **LabVIEW test bench** for T214 electronic boards in three-phase electricity meters; used Proteus simulation and UART communication for real-time data acquisition, measurement validation and fault detection, **reducing verification time by 30%**.

Projects

AI Platform for Precision Agriculture – *AgroNexus*

- Developed a precision agriculture platform for plant disease diagnosis, fertilizer recommendations, crop rotation and yield forecasting, combining **MobileNetV2/Xception, CatBoost and LSTM** models with Django REST APIs.

AI Agent for Application Screening – *AI-Driven HR Platform*

- Built an AI recruitment system combining **OCR/NLP** for CV extraction, **Sentence-BERT and FAISS** for semantic candidate matching, fine-tuned **LLaMA 3 (LoRA)** and a RAG-based chatbot.

End-to-End Telecom ML Pipeline – *Churn Prediction & MLOps*

- Built and benchmarked churn prediction models (Random Forest, XGBoost, SVM, Deep Neural Networks); automated experiment tracking and deployment with **MLflow, Docker and GitHub Actions**, exposing FastAPI inference endpoints and monitoring with Elasticsearch/Kibana.

Pizza Sales Dashboard – *Power BI Sales Analysis*

- Created an interactive Power BI dashboard using **Power Query, star-schema data modelling and DAX** to analyze sales trends, product performance and profit margins.

Technical Skills

Programming & Databases: Python, R, SQL, Java, PHP, C, MySQL, MongoDB, Oracle
AI & Machine Learning: Scikit-Learn, TensorFlow, Keras, PyTorch, XGBoost, LightGBM, CatBoost, NLP, Hugging Face, LLMs, RAG, LangChain, LangGraph, Sentence-Transformers, FAISS, LSTM
Big Data & Development: PySpark, Spark, Hadoop, Django, FastAPI, React, Spring Boot, Symfony, .NET, Angular, JavaFX
MLOps & Visualization: Docker, Jenkins, MLflow, Airflow, Git, GitHub Actions, Power BI, Grafana, Elasticsearch, Kibana, Ollama
Methodologies: CRISP-DM, TDSP, Agile (Scrum), UML

Certifications

NVIDIA: Fundamentals of Deep Learning (2025) | Applications of AI for Anomaly Detection (2025) | Building Transformer-Based NLP Applications (2025)
Oracle: OCI 2025 Certified AI Foundations Associate